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Code No: 135SC

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, September - 2021
OPERATING SYSTEMS
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1. Describe the difference between short-term, medium-term and long term scheduling. [15]

2. Explain the different types of operating system interfaces and operations in detail. [15]

3. For the following scheduling algorithms draw the Gantt chart and calculate average waiting time and turnaround time in each of the following algorithms.
   a) First-come, First-Served:

Process:	P1	P2	P3
Burst Time:	24	3	3

   b) Shortest-Job-First:

Process:	P1	P2	P3	P4
Burst Time:	6	8	7	3

   c) Priority Scheduling:

Process:	P1	P2	P3	P4	P5
Burst Time:	10	1	2	1	5
Priority:	3	1	4	5	2

   [15]

4. What is a critical section problem? How it can be solved for process synchronization? [15]

5. Define cylinder, seek time, Latency time, Transfer time. Discuss FCFS and shortest seek time first disk scheduling algorithms. [15]

6. Differentiate paging and segmentation. Give the criteria for allocation of frames. [15]

7. What is an access matrix? How the protection and security is achieved with access matrix? [15]

8. Write short notes on:
   a) fork()
   b) exec()
   c) wait() [5+5+5]

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